Client Change Request (CCR)

CCR Number: CCR-012

Client: Insight Market Research Group

Submitted By: Sarah Mitchell, Survey Operations Manager

Priority: High

Requested Delivery: End of Today's Sprint

Assigned Developer: _____

Date Assigned: _____

Business Background

Insight Market Research Group collects customer satisfaction surveys for businesses throughout the country. Some clients receive only a handful of survey responses each day, while others receive thousands. Because the number of responses varies significantly from one survey to another, the company cannot predict how much storage will be required before data collection begins.

Currently, employees estimate the number of responses that will be received, often reserving either too much or too little storage. Reserving too little storage requires employees to restart the process, while reserving too much wastes computer resources.

The Survey Operations Manager would like a computer program that asks how many survey responses will be entered and then reserves exactly the amount of storage needed. After all survey scores have been entered, the program should summarize the results and release any memory that was used.

At this time, the program only needs to process one survey during each execution.

Customer Requirements

When the program begins, it should ask the employee how many survey scores will be entered.

The program should then reserve enough storage to hold exactly that number of survey scores.

The employee should enter each survey score.

After all survey scores have been entered, the program should display:

- The highest survey score.
- The lowest survey score.
- The average survey score.

When all processing has been completed, the program should return any memory that it no longer needs before terminating.

The report should be clearly formatted and easy to read.

Acceptance Criteria

The customer will consider the software complete when all of the following are true.

□ Employee can specify how many survey scores will be entered.

□ Program reserves storage for exactly that many survey scores.

□ Employee can enter every survey score.

□ Program correctly determines the highest score.

□ Program correctly determines the lowest score.

□ Program correctly calculates the average score.

□ Program displays a neatly formatted summary report.

□ Program releases all dynamically allocated memory before terminating.

□ Program completes without errors.

Out of Scope

The customer is **not** requesting:

- Saving survey results to a file
- Reading survey data from a file
- Processing multiple surveys
- Displaying charts or graphs
- Sorting survey scores
- Determining the median or standard deviation
- Survey comments or written responses
- Databases
- Graphical user interfaces (GUI)

If these features appear in the software, they will not be evaluated.

Example Scenario 1

Employee Input

Number of Survey Scores: 5

Score 1: 85

Score 2: 92

Score 3: 78

Score 4: 96

Score 5: 89

Expected Output

Insight Market Research Group

Survey Summary Report

Survey Responses : 5

Highest Score : 96

Lowest Score : 78

Average Score : 88.00

Memory released successfully.

Program completed.

Example Scenario 2

Employee Input

Number of Survey Scores: 3

Score 1: 100

Score 2: 95

Score 3: 98

Expected Output

Insight Market Research Group

Survey Summary Report

Survey Responses : 3

Highest Score : 100

Lowest Score : 95

Average Score : 97.67

Memory released successfully.

Program completed.

Example Scenario 3

Employee Input

Number of Survey Scores: 1

Score 1: 87

Expected Output

Insight Market Research Group

Survey Summary Report

Survey Responses : 1

Highest Score : 87

Lowest Score : 87

Average Score : 87.00

Memory released successfully.

Program completed.

Notes from the Customer

"Every survey is different. One client may have only a few responses, while another may have thousands. I don't want our employees guessing how much storage they'll need before the survey begins. The program should simply ask how many responses are expected, reserve only the amount of storage that is needed, and clean up after itself when it's finished."

Developer Notes

This section intentionally left blank.

Students will complete the software requirements during class before beginning implementation.

Questions for the Customer

Before writing any code, list any questions you would ask the customer to clarify the request.

Examples:

1.

Reminder

The customer has intentionally described the problem using **business language**, not programming terminology. During class, your first task is to translate these customer requirements into **software requirements** before writing any C code.

For this assignment, you should use only the C language features that have been covered through **Lesson 14**, including:

- Variables and constants
- Input/output
- Functions
- Arrays and pointers
- Strings
- Command-line arguments
- Dynamic memory allocation (malloc(), calloc(), free())
- Basic error checking for memory allocation

Do **not** use C language features that have not yet been introduced in the course.